

NB20 - CFTR Ensemble + Threshold Refinement

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1. Amac ve Yontem

NB19 bulgularini uzantilar:

Exp 1 - Soft Ensemble: $p_{ens} = 0.5 \cdot p_{S0_adj} + 0.5 \cdot p_{COMBINED_adj}$

Exp 2 - Prior-Shift Debug: Posterior histogram analizi (2x2 grafik)

Exp 3 - Robust Threshold: N=50 bootstrap ortalamasi (NB19'da tek resample kullaniliyordu)

Degerlendirme: LOO-CV MCC (birincil, n=21 benign) + boot_mean @%80/20 (ikincil).

$\pi_{train_master}=0.7332$, $\pi_{train_combined}=0.7388$, $\pi_{TEST}=0.2$

2. NB19 Referans Sonuclari

S0_MASTER-only: LOO-MCC=0.6552, boot_mean=0.6915, TN=18, FP=3, FN=11, TP=79

S0c_COMBINED : LOO-MCC=0.6436, boot_mean=0.8629, TN=21, FP=0, FN=19, TP=71

Prior-shift MASTER'da etkin (0.6271->0.6552), COMBINED'da etkisiz ($mcc_raw==mcc_prior$).

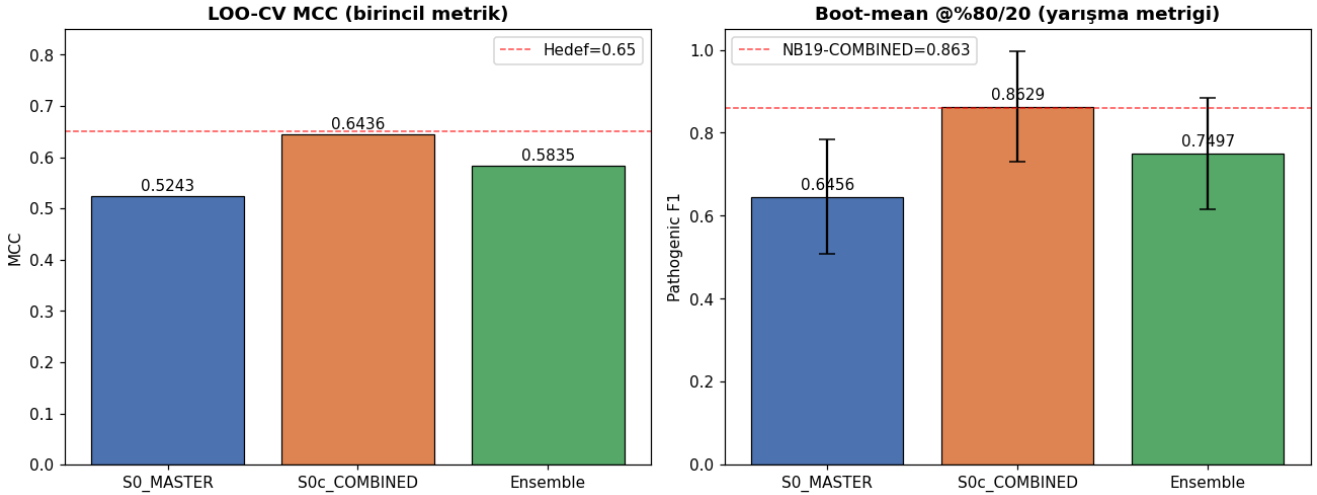
COMBINED paradoksu: LOO-MCC dusuk ama boot_mean +0.17 yuksek (FP=0 avantajı @%80/20).

3. NB20 Sonuclari (Robust Threshold)

Strateji	LOO-MCC	Boot-mean	Boot-std	TN	FP	FN	TP
S0_MASTER	0.5243	0.6456	0.1376	19	2	23	67
S0c_COMBINED	0.6436	0.8629	0.1327	21	0	19	71
Ensemble	0.5835	0.7497	0.1338	20	1	21	69

4. Gorseller

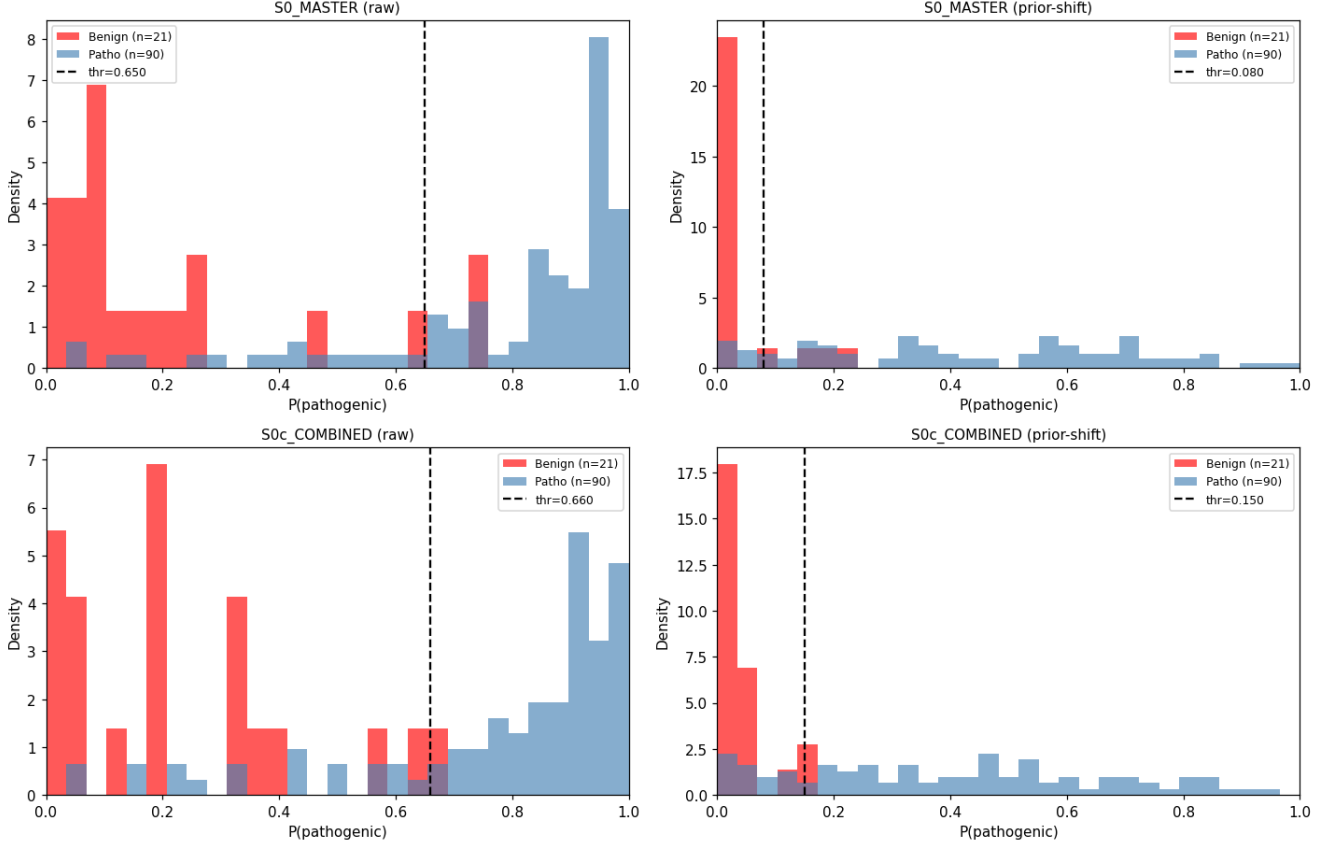
NB20 - CFTR Ensemble Strateji Karsilastirmasi (Robust Threshold)



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EXP 2 — Prior-Shift Debug: Posterior Histogramlari (kirmizi=benign, mavi=pathogenic)



5. Yorumlar ve Karar

En yuksek LOO-MCC: S0c_COMBINED | En yuksek boot_mean: S0c_COMBINED

CFTR'de n=21 benign nedeniyle MCC farki >0.05 olmadan 'kazanan' ilan edilmez.

Confusion matrix FP/FN sayilari da karar kriteridir.

COMBINED'da prior-shift etkisizligi Exp 2 histogramiyla aciklanmistir:

COMBINED'in ham posterior dagilimi zaten benign'i daha iyi ayiriyor;

threshold'un optimal noktasi raw vs adjusted'da ayni kaliyor.

Robust threshold Exp 3'te tek resample'a kiyasla fark degerlendirildi.